

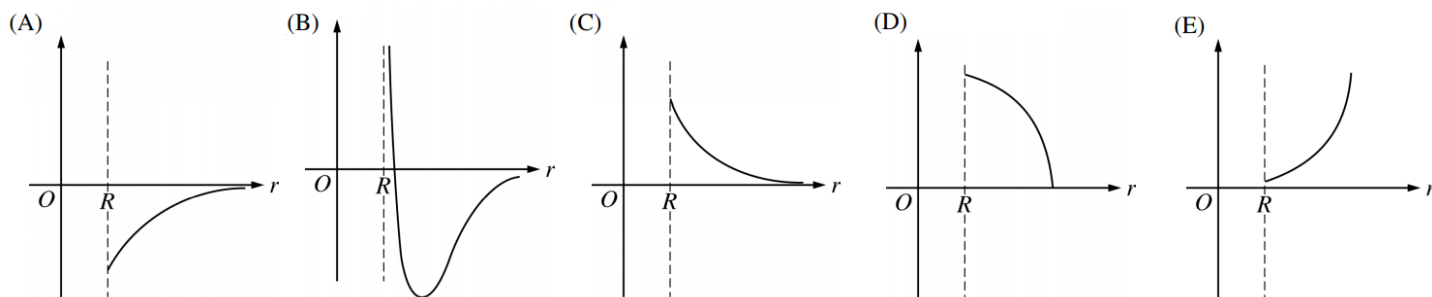


Unit 7: Gravity Quest

1. A vehicle lands on Mars and explores its surface. The average gravitational field on the surface of Mars is 3.7 m/s^2 . The weight of the vehicle is defined as the gravitational force exerted on it. Which of the following statements are true about the vehicle's weight? Select two answers.

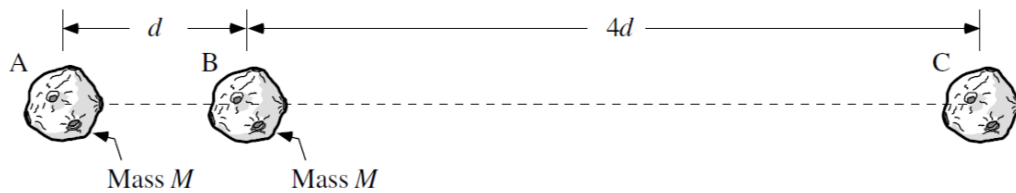
- ____ A) The vehicle's weight was constant until it reached the surface of Mars.
- ____ B) The vehicle's weight increased while it was descending to the surface of Mars.
- ____ C) The vehicle's weight always equals the normal force exerted by Mars on the vehicle while it is landing.
- ____ D) The vehicle weighs less on the surface of Mars than on the surface of Earth.

2. Ponder the graphs below, which plot some quantity as a function of distance from the center of a planet, r .



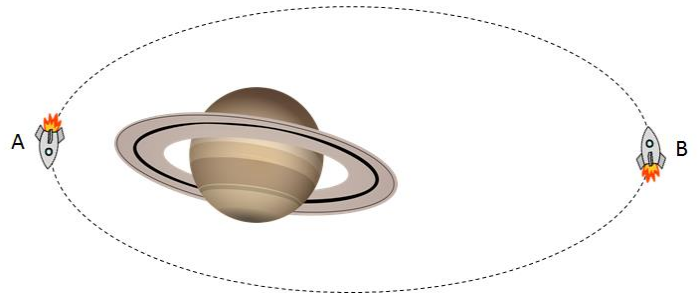
- ____ Which graph represent the force of gravity and object feels as it gets further from a planet?
- ____ Which graph represent the gravitational potential energy of the object planet system as it gets further?

3. After their defeat at the Battle of Hoth, the rebels fled the ice world to flee from the Empire's wrath. While Han and Leia went to Cloud City to fix the Millennium Falcon's hyperdrive, Luke Skywalker went to the swampy planet of Dagobah to learn the ways of the force from the wise Muppet, Yoda. The planet Dagobah has twice the radius as Earth, but only one-fourth the density. If Luke weighs 700 N on Earth, how much would he weigh on Dagobah?



4. At the instant shown, three asteroids are in a line, and the distance between B and C is 4 times as large as the distance between A and B. Asteroids A and B have the same mass. There is no net force on asteroid B due to the other two asteroids. A student makes the following comment about the mass of asteroid C:
 “Since C is four times as far from B as A is, it is only going to have one-quarter the effect on B. To get the forces on B to balance, you’d need the mass of C to be four times as large.”
 What, if anything, is wrong with the student’s statement? If nothing is wrong, state that explicitly, and explain why it is correct. If the statement is incorrect, state what is wrong and how you would correct it.

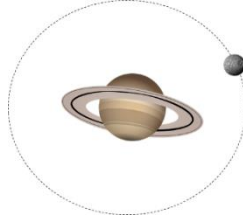
5. Lauren flies in a rocket ship around Saturn in an elliptical orbit as shown. How does her kinetic energy, potential energy, and total mechanical energy change as she flies from A to B. You do not need to justify your answer.



6. A rocket is launched from the surface of Earth and reaches a height of $5R$ above Earth’s surface. Determine an expression for the rocket’s launch velocity.

7. Saturn's moon Mimas has a mass of 1.3×10^{23} kg and orbits Saturn (which has a mass of 5.7×10^{26} kg) at a distance of 1.2×10^9 m in a circular orbit.

a) On the diagram below, draw all and label forces acting on Titan as it orbits Saturn.



b) In one revolution of Titan, does gravity due to Saturn do positive, negative, or zero work? Justify your answer.

c) Starting from Newton's 2nd law, calculate the orbital speed of Titan

d) Calculate the period of Mimas in Earth days.

e) Lando has a mass of M and a weight of W and stands on the surface of Titan. Suppose he goes somewhere where his distance to the surface of Titan triples.

i. What is Lando's mass at the new distance?

ii. What is Lando's weight at the new distance?

8. After the destruction of the first Death Star, the rebel alliance relocated their base to the planet Hoth, a remote ice planet unknown to the evil Galactic Empire. Of course, the secrecy of their hidden base would be short lived as an Imperial probe droid would locate the base, prompting Darth Vader's fleet to attack the planet. Oh, and Hoth happens to have six times the mass of Earth and twice the radius of Earth.

a) What is the approximate acceleration due to gravity on Hoth?

b) Lando, snowboards down a frictionless incline on Hoth. The incline is 250 m long and is inclined at 30° as shown. Find the speed of Lando at the bottom of the incline.

